

KIE1008 Programming II

2025/2026 Semester I

Group Assignment

Develop a **Smart Library Management System (SLMS)** to manages *books, users, borrowing records, and due dates*. The program must utilize **object-oriented programming (OOP)** principles and **advanced data structures** (such as linked lists and queue) to handle and organize information efficiently to ensure a smooth operation of a library.

Project Requirements

1. Class Structure Using Inheritance and Polymorphism

- **Base Class:** Person
 - Common attributes: name, ID, email
- **Derived Classes:**
 - Librarian — can manage books and users.
 - Member — can borrow and return books.
 - Guest — can search for books only.

2. Data Structures for Managing Records

- **Linked List** for managing *book borrowing history* (each node contains borrowing info).
- **Queue** to manage book reservation requests.

3. Library Management Interface

The system should provide a menu-driven interface with the following operations:

- Add, view, and remove books.
- Register and remove members.
- Borrow and return books (with validation).
- Search books by title/author/genre.
- Display all borrowed books and due dates.
- Fine calculation for overdue returns.

Additional Requirements

1. File I/O for Data Persistence

Implement *save* and *load* features using text or binary files:

- Save and load library records, users, and borrowing history.

2. Exception & Error Handling

Handle:

- Borrowing non-existent or already borrowed books.
- Returning books that weren't borrowed.

- Invalid user or book IDs.

Deliverables

1. **Source Code:** Organized and well-documented.
2. **Report:** A report describing how inheritance, polymorphism, and data structures were implemented, along with any challenges faced and solutions. State as well the type of contribution of each group member made to project.
3. **Video:** A video showing the operation of the SLMS, how it can be used by various type of users (librarian, user, guest), showcasing/highlighting the features that you have implemented. Text captions or voice-over should be added to the video to help with the explanation.

Form a group maximum of 5 people. Upload the video to any online video sharing platform, and include the link to the video in the report. Submit the source code and the report on the last day of Week 14 (by 5pm 23rd January 2026) through Spectrum.

Criteria	Marks				
	1	2	3	4	5
Depth of knowledge	Involves very limited fundamental engineering knowledge in problem solving.	Involves some fundamental engineering knowledge in problem solving.	Involves little knowledge that supports engineering design and knowledge of engineering technology in problem solving.	Involves some knowledge that supports engineering design and knowledge of engineering technology in problem solving.	Involves a lot of knowledge that supports engineering design and knowledge of engineering technology in problem solving, with supporting datasheets/documents.
Depth of analysis and originality	Codes are purely copied and pasted from other resources.	Little evidence of original thinking.	Some evidence of new ideas and showed originality based on someone else's idea.	Good level of originality is demonstrated based on some degree of analysis to formulate suitable model.	High level of originality is demonstrated based on extensive analysis to formulate suitable model.
Organization & Structure	Program is not possible to understand due to absence of structure.	Program is difficult to follow due to inconsistent topical shifts and jumps.	Most information is presented in logical order which is easy to follow.	All information is presented in a logical sequence but with little mess here and there.	Program is stylistically well designed and all information is presented in a logical, interesting sequence, which is easily followed.
Specifications	The program is producing incorrect results.	The program produces correct results but does not display them correctly.	The program produces correct results but does not display all of it correctly.	The program works and produces the correct results and displays them correctly.	The program works and meets all of the specifications.